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Course Code: CST8921 – Cloud Industry Trends

Lab Number: Lab 4

Lab Title: Azure Data Lake Analytics Pipeline

Submission Date: 6/8/2026

1. Introduction

In this lab, I worked with Azure Data Lake Storage Gen2 and Apache Spark to process e-commerce Parquet data. The original lab was designed to use Azure Databricks or Azure Synapse, but because there were access issues with the Azure subscription, I used a local Spark environment in VS Code as an approved alternative. The main goal was still the same: upload raw data to Azure storage, read the data with Spark, clean and transform it, write the transformed data to a refined storage area, and analyze the final results.

The dataset included customer, order, and order event files. For the main transformation, I used the `order_events.parquet` file because it contained event information such as `event_id`, `order_id`, `event_time`, and `event_type`. I connected local PySpark to Azure Data Lake Storage Gen2 using the Azure Hadoop connector, read the raw Parquet file, removed duplicate records, fixed the timestamp format, added Year and Month columns, and saved the transformed data back to the refined container.

2. Lab Walkthrough with Screenshots

1. Storage account showing Hierarchical namespace enabled

Microsoft Azure

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Home > Storage center | Blob Storage

Create a storage account

BasicsAdvancedNetworkingData protectionSecurityEncryptionTagsReview + create

[View automation template](#)

Basics

Subscription

MOC DS - 10278

Resource group

rg-cst8921-lab4-datalake

Location

Canada Central

Storage account name

cst8921lab4ilyasxxx

Primary service

Azure Blob Storage or Azure Data Lake Storage

Performance

Standard

Replication

Locally redundant storage (LRS)

Advanced

Enable hierarchical namespace

Enabled

Enable SFTP

Disabled

Enable network file system v3

Disabled

Previous

Next

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2. Container with folders

Microsoft Azure

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Home > Resource Manager | Resource groups > rg-cst8921-lab4-datalake > cst8921lab4ilyasxxx

cst8921lab4ilyasxxx | Containers

Storage account

Search

+ Add container

Upload

Refresh

Delete

Change access level

Restore containers

Edit columns

Search containers by prefix

Only show active containers

Showing all 3 items

☐	Name	Last modified	Anonymous access level	Lease state	Storage connector
☐	\$logs	6/8/2026, 7:46:17 PM	Private	Available	-
☐	raw	6/8/2026, 7:49:23 PM	Private	Available	-
☐	refined	6/8/2026, 7:49:32 PM	Private	Available	-

Containers

Classic file shares

Queues

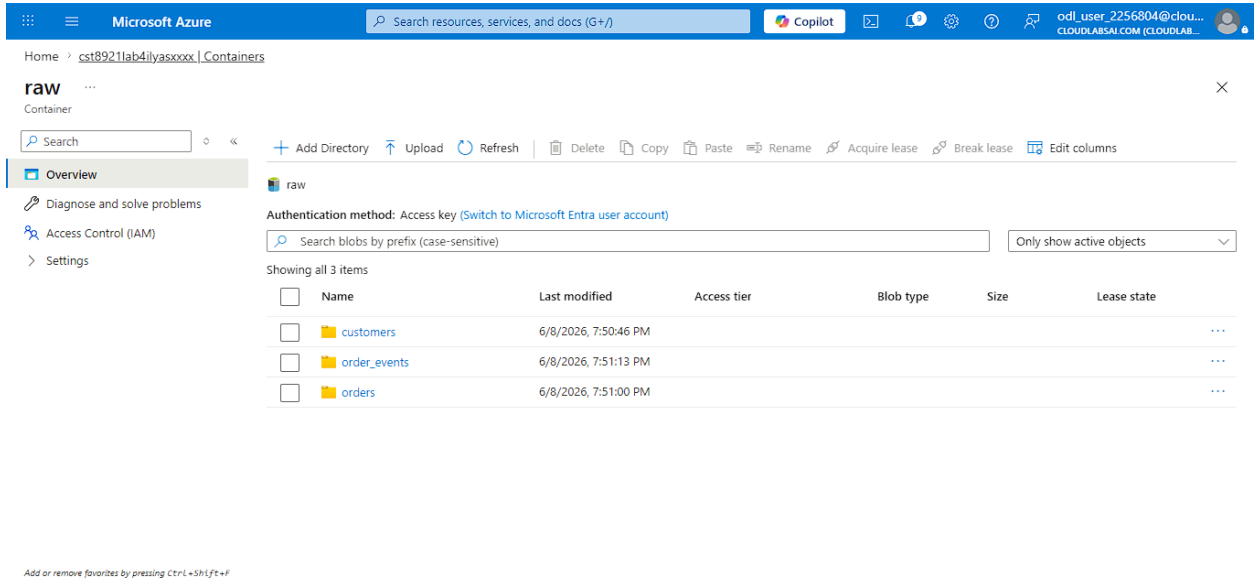
Tables

Security + networking

Data management

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3. Raw Parquet data added



Microsoft Azure

Home > cst18921lab4ilyasxxx | Containers

raw

Container

Search

+ Add Directory ↑ Upload ↻ Refresh | 🗑 Delete 📄 Copy 📄 Paste 🔄 Rename ⚙ Acquire lease ⚙ Break lease 📄 Edit columns

Overview

Diagnose and solve problems

Access Control (IAM)

Settings

Authentication method: Access key (Switch to Microsoft Entra user account)

Search blobs by prefix (case-sensitive)

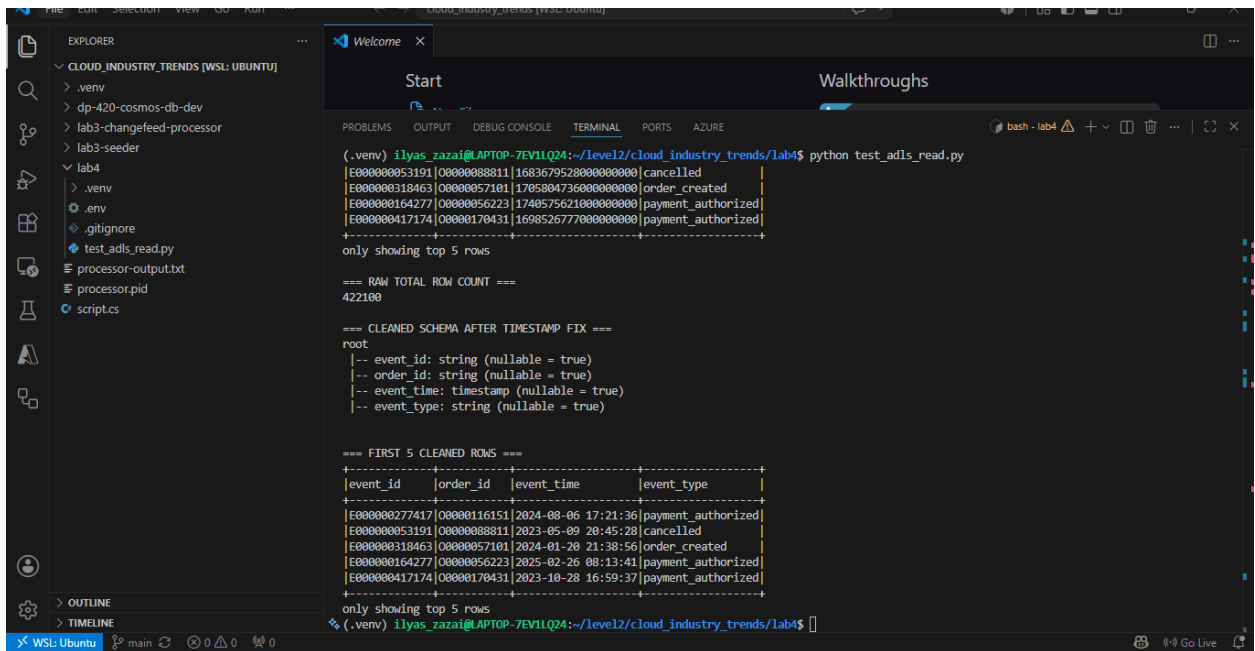
Only show active objects

Showing all 3 items

☐	Name	Last modified	Access tier	Blob type	Size	Lease state
☐	customers	6/8/2026, 7:50:46 PM				...
☐	order_events	6/8/2026, 7:51:13 PM				...
☐	orders	6/8/2026, 7:51:00 PM				...

Add or remove favorites by pressing Ctrl+Shift+F

4. Local Spark successfully reads ADLS Gen2 Parquet



```
(.venv) ilyas_zazai@LAPTOP-7EV11Q24:~/level2/cloud_industry_trends/lab4$ python test_adls_read.py
[E000000053191|00000088811|1683679528000000000|cancelled
[E0000000318463|00000057101|1705804736000000000|order_created
[E0000000164277|00000056223|1740575621000000000|payment_authorized
[E0000000417174|00000170431|1698526777000000000|payment_authorized
+-----+
only showing top 5 rows

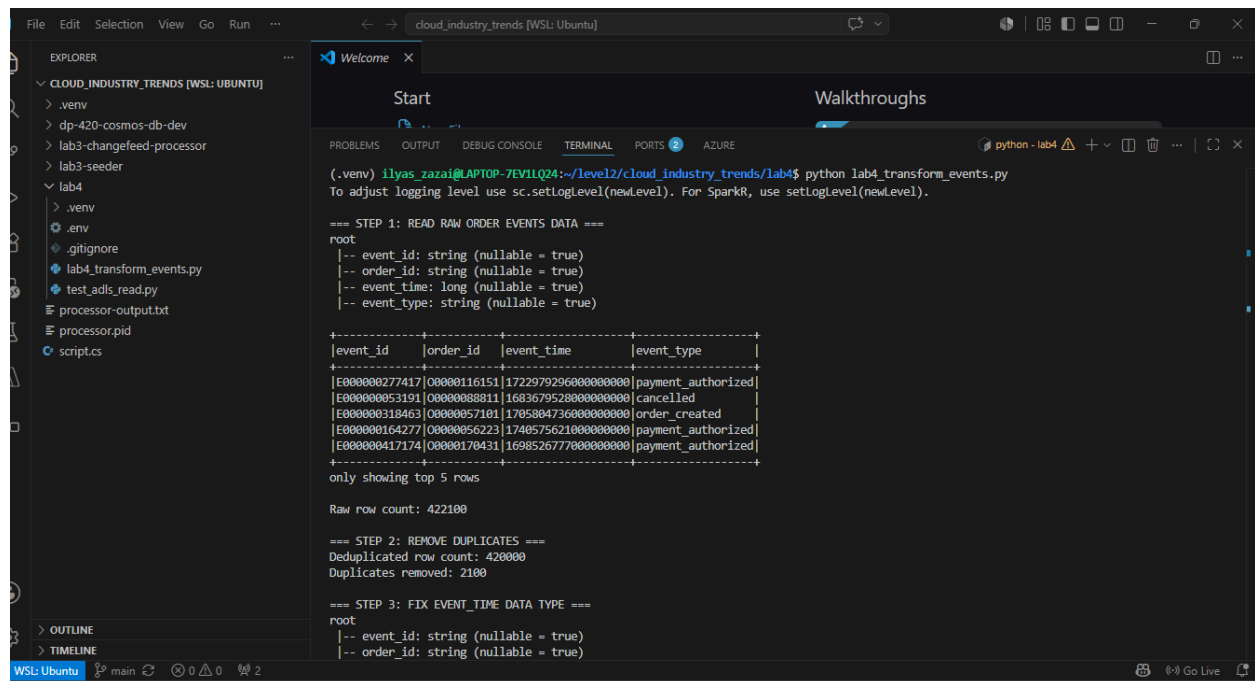
=== RAW TOTAL ROW COUNT ===
422100

=== CLEANED SCHEMA AFTER TIMESTAMP FIX ===
root
|-- event_id: string (nullable = true)
|-- order_id: string (nullable = true)
|-- event_time: timestamp (nullable = true)
|-- event_type: string (nullable = true)

=== FIRST 5 CLEANED ROWS ===
+-----+-----+-----+-----+
|event_id|order_id|event_time|event_type|
+-----+-----+-----+-----+
|E000000277417|00000116151|2024-08-06 17:21:36|payment_authorized|
|E000000053191|00000088811|2023-05-09 20:45:28|cancelled|
|E0000000318463|00000057101|2024-01-20 21:38:56|order_created|
|E0000000164277|00000056223|2025-02-26 08:13:41|payment_authorized|
|E0000000417174|00000170431|2023-10-28 16:59:37|payment_authorized|
+-----+-----+-----+-----+
only showing top 5 rows

(.venv) ilyas_zazai@LAPTOP-7EV11Q24:~/level2/cloud_industry_trends/lab4$
```

5. Raw schema, sample rows, and raw row count



```
File Edit Selection View Go Run ... cloud_industry_trends [WSL: Ubuntu]

EXPLORER
  CLOUD_INDUSTRY_TRENDS [WSL: UBUNTU]
    .env
    dp-420-cosmos-db-dev
    lab3-changefeed-processor
    lab3-seeder
    lab4
      .env
      .gitignore
      lab4_transform_events.py
      test_adls_read.py
      processor-output.txt
      processor.pid
      script.cs

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS AZURE
python - lab4

(.env) ilyas_zazai@LAPTOP-7EV11Q24:~/level2/cloud_industry_trends/lab4$ python lab4_transform_events.py
To adjust logging level use sc.setLogLevel(newLevel). For SparkR, use setLogLevel(newLevel).

=== STEP 1: READ RAW ORDER EVENTS DATA ===
root
|-- event_id: string (nullable = true)
|-- order_id: string (nullable = true)
|-- event_time: long (nullable = true)
|-- event_type: string (nullable = true)

+-----+-----+-----+-----+
|event_id|order_id|event_time|event_type|
+-----+-----+-----+-----+
|E000000277417|00000116151|1722979296000000000|payment_authorized|
|E000000053191|00000088811|1683679528000000000|cancelled|
|E000000318463|00000057101|1705804736000000000|order_created|
|E000000164277|00000056223|1740575621000000000|payment_authorized|
|E000000417174|00000170431|1698526777000000000|payment_authorized|
+-----+-----+-----+-----+

only showing top 5 rows

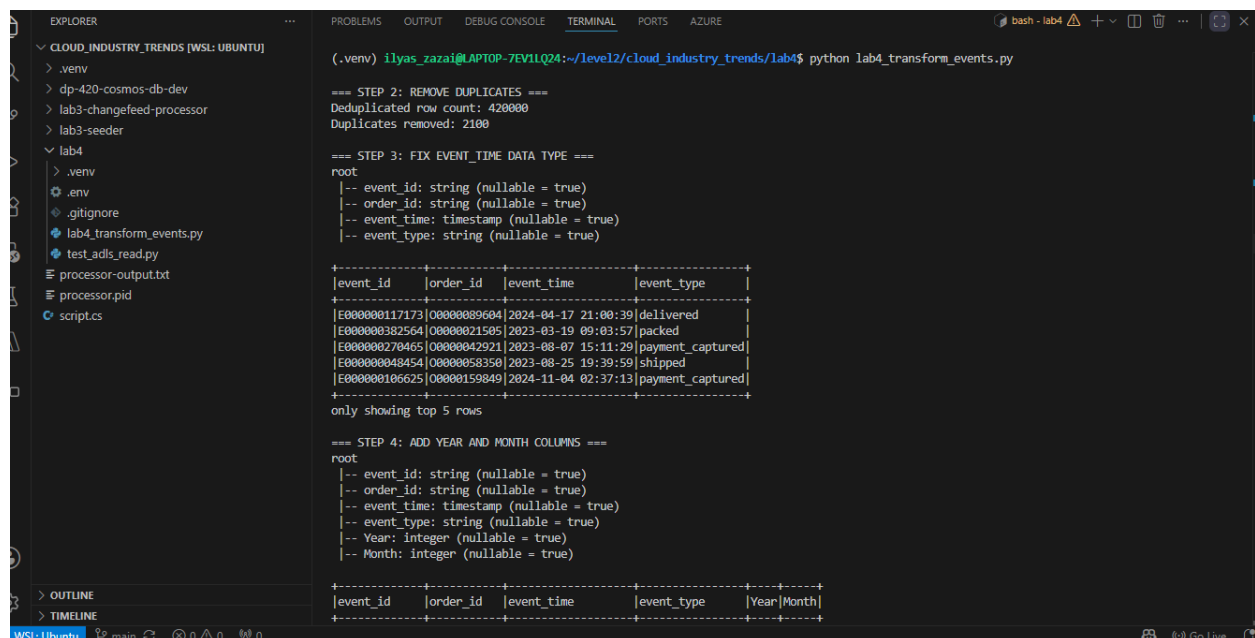
Raw row count: 422100

=== STEP 2: REMOVE DUPLICATES ===
Deduplicated row count: 420000
Duplicates removed: 2100

=== STEP 3: FIX EVENT_TIME DATA TYPE ===
root
|-- event_id: string (nullable = true)
|-- order_id: string (nullable = true)
```

6. Deduplication count showing duplicates removed

7. Clean schema with event_time timestamp, Year, and Month columns



```
EXPLORER
  CLOUD_INDUSTRY_TRENDS [WSL: UBUNTU]
    .env
    dp-420-cosmos-db-dev
    lab3-changefeed-processor
    lab3-seeder
    lab4
      .env
      .gitignore
      lab4_transform_events.py
      test_adls_read.py
      processor-output.txt
      processor.pid
      script.cs

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS AZURE
bash - lab4

(.env) ilyas_zazai@LAPTOP-7EV11Q24:~/level2/cloud_industry_trends/lab4$ python lab4_transform_events.py

=== STEP 2: REMOVE DUPLICATES ===
Deduplicated row count: 420000
Duplicates removed: 2100

=== STEP 3: FIX EVENT_TIME DATA TYPE ===
root
|-- event_id: string (nullable = true)
|-- order_id: string (nullable = true)
|-- event_time: timestamp (nullable = true)
|-- event_type: string (nullable = true)

+-----+-----+-----+-----+
|event_id|order_id|event_time|event_type|
+-----+-----+-----+-----+
|E000000117173|00000089604|2024-04-17 21:00:39|delivered|
|E000000382564|00000021505|2023-03-19 09:03:57|packed|
|E000000270465|00000042921|2023-08-07 15:11:29|payment_captured|
|E000000048454|00000058350|2023-08-25 19:39:59|shipped|
|E000000106625|00000159849|2024-11-04 02:37:13|payment_captured|
+-----+-----+-----+-----+

only showing top 5 rows

=== STEP 4: ADD YEAR AND MONTH COLUMNS ===
root
|-- event_id: string (nullable = true)
|-- order_id: string (nullable = true)
|-- event_time: timestamp (nullable = true)
|-- event_type: string (nullable = true)
|-- Year: integer (nullable = true)
|-- Month: integer (nullable = true)

+-----+-----+-----+-----+-----+
|event_id|order_id|event_time|event_type|Year|Month|
+-----+-----+-----+-----+-----+
|E000000117173|00000089604|2024-04-17 21:00:39|delivered|2024|4|
|E000000382564|00000021505|2023-03-19 09:03:57|packed|2023|3|
|E000000270465|00000042921|2023-08-07 15:11:29|payment_captured|2023|8|
|E000000048454|00000058350|2023-08-25 19:39:59|shipped|2023|8|
|E000000106625|00000159849|2024-11-04 02:37:13|payment_captured|2024|11|
+-----+-----+-----+-----+-----+
```

```
(.venv) ilyas_zazai@LAPTOP-7EV11Q24:~/level2/cloud_industry_trends/lab4$ python lab4_transform_events.py

=== STEP 6: ANALYSIS - TOTAL EVENTS BY YEAR ===
+-----+
|Year| count|
+-----+
|2023|138601|
|2024|140316|
|2025|139870|
|2026| 1213|
+-----+

=== STEP 7: ANALYSIS - EVENT COUNTS BY TYPE ===
+-----+
|event_type| count|
+-----+
|cancelled| 20931|
|delivered| 84129|
|order_created| 50479|
|packed| 58888|
|payment_authorized| 50472|
|payment_captured| 75669|
|returned| 12552|
|shipped| 66880|
+-----+

=== STEP 8: ANALYSIS - EVENTS BY YEAR AND EVENT TYPE ===
+-----+
|Year|event_type| count|
+-----+
|2023|cancelled| 6935 |
|2023|delivered| 27825|
|2023|order_created| 16611|
|2023|packed| 19565|
|2023|payment_authorized| 16695|
|2023|payment_captured| 25011|
|2023|returned| 4119 |
|2023|shipped| 21840|
|2024|cancelled| 7021 |
|2024|delivered| 28088|
|2024|order_created| 16992|
|2024|packed| 19466|
|2024|payment_authorized| 16834|
|2024|payment_captured| 25324|
|2024|returned| 4271 |
|2024|shipped| 22320|
|2025|cancelled| 6918 |
|2025|delivered| 27968|
|2025|order_created| 16728|
|2025|packed| 19669|
|2025|payment_authorized| 16807|
|2025|payment_captured| 25109|
|2025|returned| 4135 |
|2025|shipped| 22536|
|2026|cancelled| 57 |
|2026|delivered| 248 |
|2026|order_created| 148 |
|2026|packed| 188 |
|2026|payment_authorized| 136 |
|2026|payment_captured| 225 |
```

8. Analysis Event by years and event type

```
(.venv) ilyas_zazai@LAPTOP-7EV11Q24:~/level2/cloud_industry_trends/lab4$ python lab4_transform_events.py

=== STEP 8: ANALYSIS - EVENTS BY YEAR AND EVENT TYPE ===
+-----+
|Year|event_type| count|
+-----+
|2023|cancelled| 6935 |
|2023|delivered| 27825|
|2023|order_created| 16611|
|2023|packed| 19565|
|2023|payment_authorized| 16695|
|2023|payment_captured| 25011|
|2023|returned| 4119 |
|2023|shipped| 21840|
|2024|cancelled| 7021 |
|2024|delivered| 28088|
|2024|order_created| 16992|
|2024|packed| 19466|
|2024|payment_authorized| 16834|
|2024|payment_captured| 25324|
|2024|returned| 4271 |
|2024|shipped| 22320|
|2025|cancelled| 6918 |
|2025|delivered| 27968|
|2025|order_created| 16728|
|2025|packed| 19669|
|2025|payment_authorized| 16807|
|2025|payment_captured| 25109|
|2025|returned| 4135 |
|2025|shipped| 22536|
|2026|cancelled| 57 |
|2026|delivered| 248 |
|2026|order_created| 148 |
|2026|packed| 188 |
|2026|payment_authorized| 136 |
|2026|payment_captured| 225 |
```

9. Refined container showing transformed output folders

Microsoft Azure

Search resources, services, and docs (G+)

Copilot

odl_user_2256804@clou...
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Home > Storage center | Blob Storage > cst8921lab4ilyasxxx | Containers

refined
Container

Search

+ Add Directory ↑ Upload ↻ Refresh | 🗑 Delete 📄 Copy 📄 Paste ⚙ Rename ⚙ Acquire lease ⚙ Break lease 🛠 Edit columns

Overview

Diagnose and solve problems

Access Control (IAM)

Settings

refined > order_events_refined

Authentication method: Access key (Switch to Microsoft Entra user account)

Search blobs by prefix (case-sensitive)

Only show active objects

Showing all 5 items

☐	Name	Last modified	Access tier	Blob type	Size	Lease state
☐	[-]					...
☐	Year=2023	6/8/2026, 8:43:19 PM				...
☐	Year=2024	6/8/2026, 8:43:19 PM				...
☐	Year=2025	6/8/2026, 8:43:19 PM				...
☐	Year=2026	6/8/2026, 8:43:19 PM				...
☐	_SUCCESS	6/8/2026, 8:48:59 PM	Hot (Inferred)	Block blob	0	Available

Add or remove favorites by pressing Ctrl+Shift+F

11. Resource group delete command or deletion confirmation

Microsoft Azure

Search resources, services, and docs (G+)

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Resource groups

Storage accounts

Azure Synapse Analytics

Azure Cosmos DB

Policy

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Quickstart Center

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Resources

Recent Favorite

Name	Type	Last Viewed
rg-cst8921-lab4-datalake	Resource group	a few seconds ago
cst8921lab4ilyasxxx	Storage account	11 minutes ago
ODL-Azure-2256804	Resource group	5 days ago

See all

Notifications

More events in the activity log → Dismiss all

Deleted resource group rg-cst8921-lab4-datalake
Deleted resource group rg-cst8921-lab4-datalake
a few seconds ago

3. Reflection

This lab helped me understand how a real data lake workflow works. I learned that raw files are first stored in a data lake, then processed with Spark, and finally saved into a refined zone for analysis. I also learned that Spark can run in different environments. Even though I could not use Synapse or Databricks because of subscription access issues, I was still able to complete the same data processing workflow using local Spark in VS Code.

One important problem I faced was the Parquet timestamp format error. The `event_time` column was stored as a nanosecond timestamp, and Spark did not read it correctly at first. I fixed this issue by changing the Spark configuration and converting the timestamp into a readable format. This was a useful troubleshooting experience because real cloud and data engineering work often includes data format and compatibility issues.

I also learned the importance of separating data into raw and refined zones. The raw zone keeps the original files unchanged, while the refined zone stores cleaned and transformed data. This is a common pattern in modern data engineering and cloud analytics.

5. Conclusion

In conclusion, this lab showed how Azure Data Lake Storage Gen2 and Apache Spark can be used together to process and analyze Parquet data. I uploaded raw data to Azure storage, connected local PySpark to the storage account, read the Parquet file, cleaned the data, added new time-based columns, and saved the transformed output to the refined container.

The final analysis showed event counts by year and event type, which helped explain the order event lifecycle in the dataset. This lab also gave me practical experience with Spark, Parquet files, ADLS Gen2, data transformation, partitioned output, and basic analytics. Even though I used local Spark instead of Synapse or Databricks, the main cloud data processing concept remained the same